POOL PLANT ROOM
AS-FOUND SURVEY

CURRENT VERSION
albidr.github.io/Pool-ADR

POOL VOLUME approx. 70 to 75 m^3	POOL TYPE Overflow + balance tank	POOL FINISH Black resin, smooth	PLANT ROOM 4.52 m^2, below ground
BALANCE TANK 5.46 m^3, next door	FILTRATION Sand, Kripsol 6-way	PUMP DAB EuroPro 200 M	SANITISATION Salt, Zodiac TRi PRO
FILTER Kripsol Ø760, 225 kg			

QUICK REFERENCE

EVERYTHING NORMAL

- M on 1, Filtration
- B and C open, A almost closed
- All four D open
- G and 7 open, E open
- L shut or barely open, H shut, 1 shut
- Gauge 4 at **0.5 to 0.7 bar**
- pH 7.2 to 7.4, chlorine 0.6 to 1.5

WATER COMING IN

- The pool is above you. **Stopping the pump does not stop the water.**
- Shut 7, then L, then all four D, then G
- Check the sump pump N is running. It is the only drain.

THREE THINGS NEVER TO DO

- Never move the Kripsol lever with the pump running. Kill the panel first.
- Never put a chlorine product near the acid drum. The drum is **sulfuric acid**, whatever the old sheet says.
- Never run the cell with no flow. Isolate on B and C.

WHERE EACH STATEMENT COMES FROM

OBSERVED

Read off the photographs: nameplates, printed labels, and pipe runs I could follow end to end. Counted against a second image wherever countable.

HOUSEHOLD REFERENCE

The four printed sheets kept in the room, written for this pool and used for years. Not a professional specification, and not necessarily complete. They are the standing house procedure and the shared vocabulary for this plant room, and they are the primary record of how it was run.

MANUFACTURER DOCUMENTATION

The makers' own manuals for the filter, the chlorinator, the TRi PRO module and the pump, downloaded and kept in *09_manuals/* with the official link for each. Where a manual and the printed sheets disagree, this document follows the manual and says so.

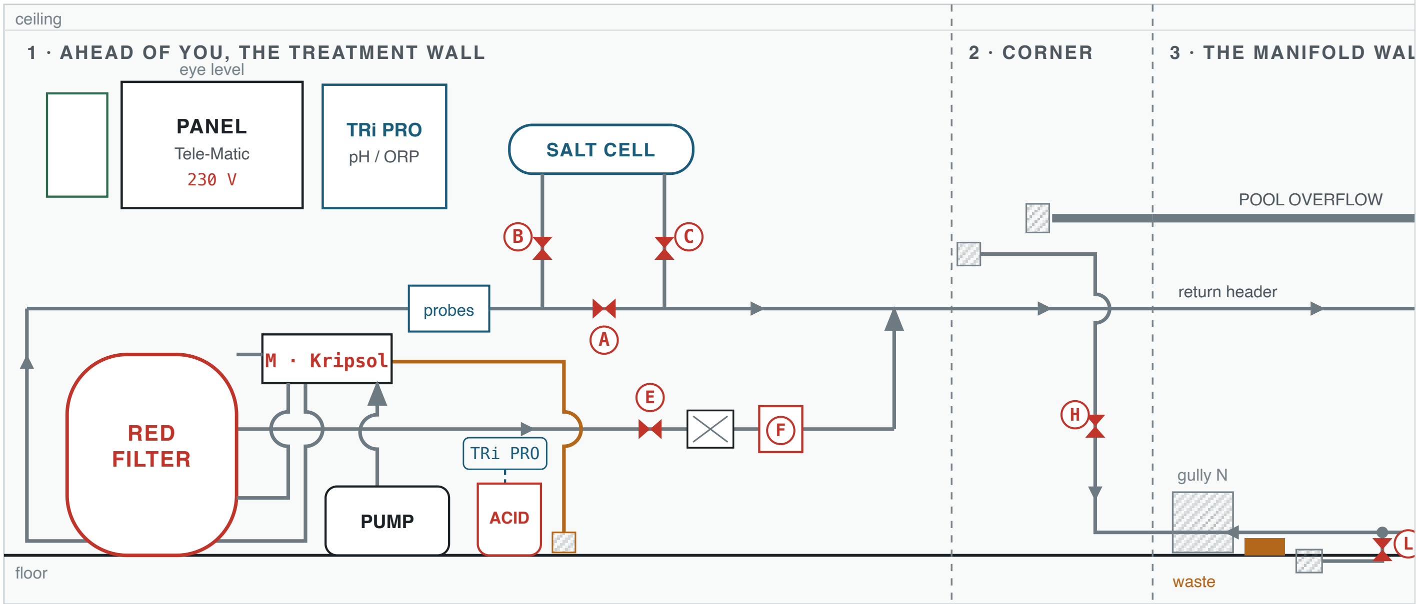
INFERRED

My reading rather than an observation. Very little is left: what remains is named as mine where it appears, most of all the isolation sequence in the safety notes, which follows from the levels and not from the sheets.

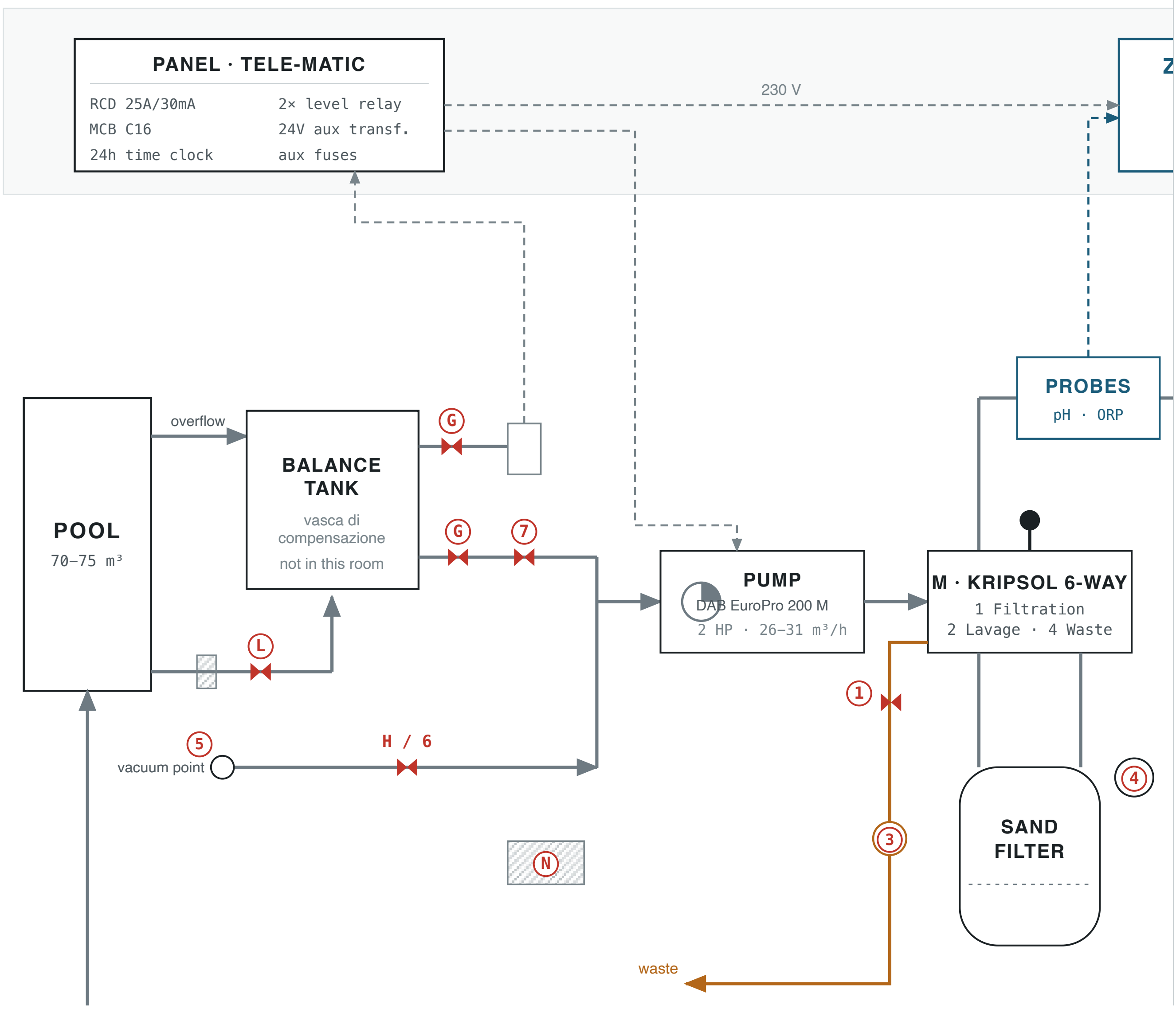
§1 THE WHOLE INSTALLATION, ONE DIAGRAM

This uses the letters and numbers already written on the printed sheets (A, B/C, D, E, F, G, H, L, M, N, and 1 to 7) rather than tags of my own, so that the drawing, the sheets and anything said out loud in the room all use one vocabulary. Every key is drawn as a circle, on the drawing and in the text alike.

STAND IN THE DOORWAY, TURN RIGHT, AND THIS IS WHAT YOU PASS



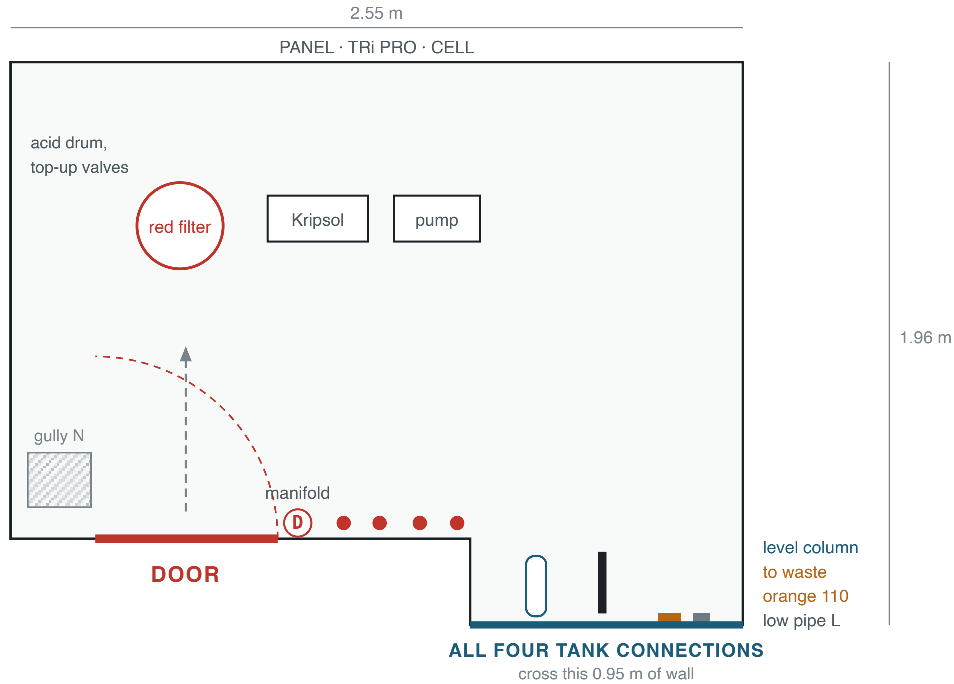
ELECTRICAL, DOSING & CONTROL



back to the pool

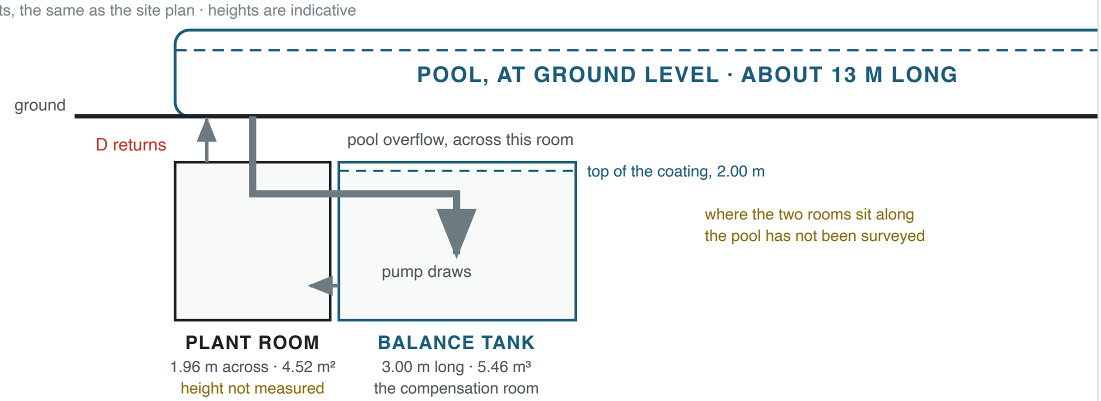
PLAN OF THE PLANT ROOM

to scale, 1 m = 200 units · 4.52 m² of floor



SECTION: HOW THE SPACES STACK UP

to scale sideways, 1 m = 55 units, the same as the site plan · heights are indicative



Five views, and two of them are now to scale. The wall strip is what you see turning right from the doorway; its order follows the 360° pan and its distances are not to scale. **The plan of the plant room is to scale** at 1 m = 200 units, drawn from your measured 2.55 by 1.96 m with the 1.60 by 0.30 m entrance notch, so 4.52 m² of floor. **The site plan is to scale for the rooms** at 1 m = 55 units: the balance tank runs out at right angles from the bottom-right corner, 0.91 by 3.00 m, coated 2.00 m up, and all four of its connections cross that one 0.95 m of wall. The service corridor on it is schematic, because its size has not been measured. The section shows the pool sitting above both underground rooms, which is why its water will find any opening down here. Every tag is explained below.

READING THE DRAWING

pipe — larger bore — to waste — electrical or signal — a valve

a small arc means two pipes cross without joining — a wall the balance tank is behind

on the site plan, schematic and not measured

THE LETTERS, AS THE SHEETS NUMBER THEM

A	Bypass around the salt cell. **Keep almost closed**: it sets how much water skips the cell.
B, C	Cell inlet and outlet. **Keep open.**
D ×4	Return valves on the manifold. **All four stay open**, or the circuit throttles and filter pressure rises.
E	Mains top-up tap. **Keep open** so the automatic top-up can work. The line runs on to the left behind the filter and **discharges into the return manifold**, so mains water reaches the pool through the four wall outlets. It does not feed the tank; the tank fills from the pool spilling over its deck channel.
F	Solenoid on the top-up, opened by the two level relays in the panel. A manual bypass loop beside it lets you fill by hand if it fails, and it is also the way to speed up a refill, because a top-up solenoid is a small orifice by design. **The bypass has no cut-out**, so close it before the pool crests or it will run on into the tank's safety overflow and out to waste.
G	Compensation valve, on the low pipe behind the return drops.
H	Manual vacuum. **Normally shut.** The port is in the pool wall: unscrew the cover, push the hose in, and the pump pulls dirt off the floor.
L	Pool floor suction, the *presa di fondo*. **Shut, or barely open** so the deep water is renewed rather than sitting still. Open fully to empty the pool.
M	Kripsol six-way valve, mounted on the filter's side. **Position 1, Filtration** for normal running.
ACID	Sulfuric acid 36%, the pH minus. A small plastic tube runs from the drum up to the TRi PRO, whose peristaltic pump draws from it and doses into the circuit; that much is confirmed on site. **Where it injects is still not established**: follow the tube leaving the TRi PRO and see which pipe it enters.
N	Floor gully and sump with a submersible pump. The room is below ground, so this is the only way water leaves the floor.
1	Filter drain valve. Open it before a backwash, close it after.
3	Sight glass. Run the backwash until the water in it comes clear.
4	Filter pressure gauge. **0.5 to 0.7 bar is normal running. Backwash once it passes 1.0.** The sheet's “about 1 bar” is the backwash trigger, not the target.
7	Balance tank outlet. Close it to vacuum harder; close it when emptying so the tank does not drain too. **Unresolved:** the sheets describe 7 and G as doing the same job, and only one valve in that role is visible in any photograph. Either they are one valve under two names, or there is a second

one on the run between the compensation valve and the pump. Follow that pipe and look for a second red handle.

PANEL SETTINGS FOR NORMAL RUNNING

pump Filtration pump on A, automatic, so the 24 hour time clock runs it.

top-up Reintegro on A, automatic, so the level relays call for water on their own. The top-up discharges into the **return manifold**, so mains water reaches the pool through the four wall outlets, not into the tank.

interlock **There is a low-level cut-out that will not let the pump run when the balance tank is too low.** The printed sheet states it plainly in the backwash procedure, "the balance tank level drops during this and will stop the pump", and it was confirmed by accident on 2026-09-07 when the pump sat on M for a full day with an empty tank and never started. Treat it as a backstop for the day someone forgets, never as a substitute for the selector being on 0: the same clear column that feeds it has its own isolating valve, and closed, that valve makes the floats read a satisfied level and the interlock **permit** a dry run.

HOW THE WATER MOVES, IN ORDER

1. The pool spills over its deck channel into the **balance tank**, which is in the compensation room next door, not in here. The big pipe crossing the manifold wall high up is that overflow passing through.
2. The pump draws from the balance tank through G and 7. The **pool floor suction** L and the **manual vacuum** H feed the same low pipe, which runs behind the four return drops and crosses them without joining.
3. The pump pushes up into the Kripsol valve M, which on position 1 sends the water into the **filter's upper union**.
4. It comes back out of the **filter's lower union** through M, drops to the floor, runs left behind the filter, up the wall, and rejoins the main run by the panel.
5. From there it passes the **probes** and the **salt cell**, with A setting how much skips the cell.
6. The **return header** takes it along the manifold wall and down the four D drops, through the floor, back to the pool.
7. On backwash, waste leaves through M along the floor and out through the wall in the **orange 110**.

NOT IN THIS ROOM

tank The balance tank runs out at right angles from the bottom-right corner of this room: 0.91 m wide, 3.00 m long, coated 2.00 m up, so 5.46 m^3 to the coating line. Its four connections come through that corner wall. Reached only by a small hatch at the far end, 3 m from the connections. Photographs in *05_balance-room/*, dimensions in *DIMENSIONS.txt*.

valve The main shut-off for the whole pool system is in the boiler room.

§2 THE TWO ROOMS, MEASURED

Measured 2026-09-07, so the drawing above and the tape now agree. The dimensioned sketch and the scaled site plan are in *06_drawings/*; the full working is in *DIMENSIONS.txt*.

	THIS PLANT ROOM	THE BALANCE TANK
`Shape`	2.55 × 1.96 m, with a 1.60 × 0.30 m notch at the entrance corner	0.91 × 3.00 m, coated 2.00 m up
`Floor`	`4.52 m²`	`2.73 m²`
`Volume`	At least 2.33 m tall from the 3D scan, so **10.5 m³ of air or more**. Not closed: the scan captured no ceiling plane.	5.46 m³ to the coating line, but the safety overflow is at 1.35 m, so **3.69 m³ is the most it ever holds**
`Getting in`	The door, bottom left on the plan.	One small hatch high in the service corridor. A ladder to climb to it and a second lowered inside to reach the floor.

THE TANK'S FOUR CONNECTIONS, ALL THROUGH ONE 0.95 M OF WALL

`in` The pool overflow. A large orange-brown pipe entering high, ending open and **lying loose on the tank floor**. Clipping it would cost nothing and stop it knocking when the pool spills hard.

`out` The tank outlet, **0.35 m above the tank floor**, through the compensation valve G. **This is the pump's entire water supply.** On an overflow pool the pump does not draw from the pool, it draws from the tank. Below 0.35 m the pump has nothing; start it at about 0.55 m so the socket has cover and does not vortex.

`level` The take-off for the clear column in this room, alongside the outlet at **0.35 m**, so the column can only read levels above that. Communicating vessels: the column stands at exactly the tank's level, so it is a remote gauge for a room you cannot see into. Its float switches drive the two level relays, which open solenoid F.

`safety` The high-level overflow to waste, **1.35 m above the tank floor**, not at the top: there is 0.65 m of coated tank above it that never holds water. No valve, no electrics, nothing to fail. If the tank fills past it, water leaves instead of rising.

FOUR NUMBERS WORTH CARRYING

`27.3 L` per centimetre of tank level. This one turns the clear column from a vague indicator into a measuring instrument.

`5.1%` of pool volume, taking the 3.69 m³ the tank can actually reach. Design practice for a deck-level pool starts at 5 to 10 percent, so **the tank is adequate rather than generous**: it sits at the floor of that band, and the working band alone is 3.8 percent, below it. The 7.5 percent you get from the coating line is not reachable in normal operation.

29 cm of level is what ten people getting in at once takes out, at the design figure of 80 litres a bather. Bather surge is not this tank's constraint.

33–55 cm **is what one backwash and rinse takes**, out of a band that is exactly 1.00 m. That is the constraint, and it is why the sheet warns the pump will stop. Ten bathers take another 29 cm, so **do not backwash with a full pool of people in it**.

WHY THE SHEET SAYS THE PUMP WILL STOP DURING A BACKWASH

- Two to three minutes of backwash at 18 to 26 m^3/h, plus the rinse, removes roughly **900 to 1,500 litres**. Against a working band of about 2,700 litres that is **a third to a half of everything the tank has to give, in one cycle**.
- So the note on the printed sheet is not vague caution. It is an accurate description of a tight margin. **Top the tank up before you backwash, not after.**
- That same water carries salt away with it: 3.6 to 6.0 kg. One 25 kg sack replaces the loss of four to seven backwashes.

BEFORE ANYONE GOES INTO THE TANK

- It is a confined space. One small opening, high up, below ground, no second way out, ladder in and ladder down. **Never alone.**
- The four connections are on the wall **3 m from the hatch**, so reaching them means traversing the full length of a chamber 0.91 m wide and 2.00 m deep.
- The lowest connection is at 0.35 m and there is no sump, so **the tank cannot be emptied through any fixed connection**: **956 litres** stay below the outlet and have to be pumped or sponged out.
- *[added]* There is rust-coloured staining running down the corridor wall below the hatch, and blown plaster at the sill. Water has been coming through or condensing there. Worth understanding while the tank is empty and open.

§3 PROCEDURES, AS WRITTEN ON THE SHEETS

Transcribed and translated from the four printed sheets, with their numbering kept. I have not changed the order of any step. Where I have added something the sheets do not say, it is marked *[added]*.

BEFORE MOVING THE KRIPSOL LEVER

- The sheets are explicit: **switch the panel off at the main switch first**, move the lever, then switch back on. Turning that valve against a running pump is what destroys the seal inside it.
- *[added]* Never run the pump with the lever between positions.
- *[added]* During backwash, rinse and waste the water goes to drain rather than back to the pool, so the cell should not be producing into the sewer.

KRIPSOL POSITION	MARKED ON THE DIAL	WHAT IT DOES
1	FILTRACION / FILTRATION	Normal running. Down through the sand, out to the returns.
2	LAVADO / BACKWASH	Reverse flow, up through the sand, out to waste.
3	ENJUAGUE / RINSE	Settles the bed after a backwash, still to waste. This is the "fare risciacquo" step.
4	DESAGUE / WASTE	Straight to drain, bypassing the filter. Used for emptying and for vacuuming heavy dirt away.
5	CIRCULACION / RECIRCULATE	Round the circuit with the filter bypassed.
6	CERRADO / CLOSED	Shut. Never run the pump on this.
0	INVIERNO / WINTER	Winter position, handle parked between seats so the gasket is not compressed all season.

Cast around the rim of the valve itself, in four languages: **NEVER SHIFT HANDLE WHILE PUMP IS RUNNING**.

BACKWASH THE FILTER (CONTROLAVAGGIO)

1. Switch off the main switch on the panel.
2. Open filter drain valve 1 (vertical).
3. Move Kripsol lever M to position 2 (Lavage).
4. Switch the panel back on.
5. Leave it running until the water in sight glass 3 runs clear.
6. Switch the panel off.
7. Return drain valve 1 to horizontal.
8. Return lever M to position 1 (Filtration).
9. Switch the panel back on.
10. Do a rinse.

Note from the sheet: the balance tank level drops during this and will stop the pump. Close valve G temporarily. Gauge 4 reads **0.5 to 0.7 bar** when the sand is clean.

VACUUM THE POOL FLOOR (ASPIRAZIONE FANGHI)

1. Attach the pole to the vacuum head.
2. Fill the hose with water as fully as you can, so it does not draw air.
3. Unscrew the cover and put the hose into outside point 5.
4. Filtration must be running, pump on.
5. Open vacuum valve 6 (vertical).
6. Close balance tank outlet 7 (transverse). It does not have to be fully closed: the more you close it the harder it sucks. If the head sticks to the floor, open it a little.
7. Reverse the process to finish.

Handwritten on the sheet: the basket under the clear lid after the pump has to be cleaned, by unscrewing the two black knobs. And: with the Kripsol lever on position 4, the dirt is discharged to drain instead of passing through the filter, but then the water you suck out is thrown away.

REFILL AND RESTART, AFTER THE SYSTEM HAS BEEN DRAINED

1. **Pump control to 0 before anything else.** Not A: on automatic the 24 hour time clock will start it on its own schedule, and the pump's seal is water-lubricated. Dry running destroys it in minutes. There is a low-level interlock that should also hold it off, but it is a backstop, not a plan.
2. **Chlorinator off.** Fresh water has no salt, and electrolysis at low salt drives the cell voltage up and damages it. It will show *Check Salt* until the salt is back, which is correct rather than a fault.
3. **Reintegro on A, all four D valves open.** The top-up discharges into the return manifold, so mains water reaches the pool through the four wall outlets. **It does not fill the tank.** On A it stops itself when the tank floats satisfy; on M you have forced the solenoid open with no cut-out.
4. **Leave the Kripsol lever alone unless the gauge says otherwise.** Kripsol's manual says to fill from the mains on 6, closed, so the water bypasses the filter. In this installation that is not needed: with the lever on 1 and the mains flowing, **the gauge reads zero**, so nothing is being over-pressurised. Check the gauge; if it shows pressure with the pump off, go to 6. Otherwise save the lever movement.
5. **Expect a day or more.** This line delivers roughly 1.5 to 2 m^3/h, so 72 m^3 takes 36 to 48 hours. To measure your own rate: one centimetre of pool level is about 400 to 500 litres, so time the rise over an hour. If it is slower than it should be, the **manual bypass loop** past the solenoid is the biggest single improvement available, and the **strainer** on the top-up line is the most likely restriction after a drain. **Close the bypass before the pool crests**, because unlike the reintegro it will not stop itself.
6. **While it fills, hold the standing water.** It has no sanitiser and no circulation. Pre-dissolve **2 to 2.5 g per m^3 of water actually in the pool** of the granular chlorine, so about 150 g at 60 m^3, and pour it round the perimeter. Repeat every two or three days. That costs about 1 mg/L of cyanuric acid, against 8 for a full shock. Write down what you added.
7. **Watch the clear column, not the pool.** The instant it starts to rise, the pool is spilling and the tank is filling. From there it is **about 45 minutes to an hour to reach 0.55 m**, your start level, and roughly another hour before the reintegro satisfies and shuts off.
8. **Kripsol's pre-start checks.** Turn the motor's fan blades by hand from the back to confirm it is not seized after standing dry. Check the thermal overload has not tripped. Open the suction and discharge valves and **fill the pump body with water by hand** through the clear strainer lid, and check that lid's O-ring, which is what stops it sucking air.
9. **Valves:** G and 7 open, all four D open, B and C open, A almost closed, L and H shut, filter drain 1 shut.
10. **First start on 5, circulation, not 1.** Kripsol's instruction, and it is not the obvious choice: circulation bypasses the filter, so you are not driving air and disturbed debris through the sand on the first run. Panel off before the lever moves, handle pressed down as a clutch. **Watch the strainer lid clear of air within a minute or two. If it is still churning, stop.**
11. Then 2 backwash until the sight glass runs clear, two to three minutes, then 3 rinse briefly, then 1 filtration. Panel off for every movement. **Write down the gauge reading after that last backwash**: it is the new clean baseline and should be 0.5 to 0.7.

12. **Salt: about 290 kg, twelve 25 kg sacks**, added with the pump running so it dissolves and circulates. Twelve sacks lands 72.5 m³ at 4.1 g/L. **Do not exceed 5.0 g/L**, the robot's ceiling. Only then switch the chlorinator on.
13. **Calibrate the pH probe** in pH 7.5 buffer. You have replaced the entire body of water, so whatever it was trimmed to no longer applies. Check the acid drum has something in it before the TRi PRO starts dosing.
14. **Keep the robot out** until free chlorine is under 4 mg/L, which rules it out during any commissioning shock.

Nothing here can flood. The pool's level is capped by the deck channel, the channel drains to the tank, and the tank has a passive high-level overflow to waste. A stuck float sends mains water quietly to the drain rather than onto a floor. *[added]* The one exception is the manual bypass, which has no cut-out at all.

EMPTY THE POOL

1. Pump off. Kripsol lever M to 4 (Waste). Valve 1 vertical (open).
2. Valve 7 horizontal (closed). If you leave it vertical (open) the balance tank empties as well.
3. Valve L open (pool floor suction). With M on 4 the pump draws off the bottom and sends it to waste, which is what empties the pool.
4. Pump control to M (manual).
5. Top-up control to 0.

TARGET	VALUE	FROM THE SHEETS
pH	`7.2 to 7.4`	Below 7 is acidic, above 7.6 is basic.
Free chlorine	`0.6 to 1.5`	1 is ideal.

Robot's water limits	`Cl max 4 mg/L`
	`pH 7.0 to 7.8`
	`6 to 35 °C`
		From the Maytronics manual. Your normal targets sit inside all three. Two places they bite: **a commissioning shock at 10 to 15 g/m³ gives about 8 mg/L, double the robot's chlorine limit, so take the robot out and leave it out until free chlorine is back under 4**; and a black resin pool in an Italian summer can approach the 35 °C ceiling. Below 15 °C its climbing suffers, which is the same threshold at which the sheets say to switch the electrolyser off.

Effect of the dark finish	`runs warm`	A black surface absorbs solar heat, so this pool sits warmer than a pale one. Three consequences worth knowing: **chlorine is consumed faster**, so the cell works harder; **evaporation is higher**, so the top-up runs more and carries away more salt; and **calcium is less soluble in warm water**, so it scales more readily, which is what the Svernante on the shelf is for. Scale also shows up starkly as white film on black resin, which is a useful early warning you would not get on a pale pool.
Filter pressure	`0.5 to 0.7 bar`	**Backwash once it passes 1.0.** Record the reading straight after a backwash: that is this filter's clean baseline, and the climb away from it is the real signal. Kripsol's own manual, correcting the sheet's “about 1 bar”.
Salt	`4.0 to 5.0 g/L`	**Narrower than the sheets say, and the reason is the robot.** The sheets give 4 to 9 g/L and Zodiac asks for above 4. But the Maytronics manual states a hard maximum of **5,000 ppm, that is 5.0 g/L**, above which the robot is out of specification. The window for running both machines is therefore 4.0 to 5.0. At 4 g/L a 72.5 m³ pool needs about **290 kg, twelve 25 kg sacks**; twelve sacks lands you at 4.1 g/L, comfortably inside both limits. Do not aim for the middle of the old 4 to 9 range.
Salt top-ups	`25 kg sacks`	Two or three at a time when the unit signals. Only modest additions are needed after the initial charge: salt is lost to backwashing, not to evaporation.
pH down dosing	`about 1 kg`	Per 0.2 of pH reduction.

TARGET	VALUE	FROM THE SHEETS
Electrolyser, water below 15 °C	`off`	Pointless: algae and bacteria growth is reduced or stopped.
Electrolyser, water above 15 °C	`on`	Needed: bacteria, fungi and algae reproduce quickly at higher temperatures.

THE ONE RULE THE SHEET PUTS IN BOLD

- **Correct pH first.** Then the others, one at a time. Never adjust two things together.

The sheets also cover: clear water with algae, shock chlorinate and brush the floor; cloudy water with algae, the same plus check the gauge and backwash; cloudy water alone, test and consider flocculant, which clumps fine particles so the filter can catch them. Higher temperature and more swimmers both call for more chlorination, either the BOOSTER function or manual product.

§5 WHEN SOMETHING GOES WRONG

None of this was on the printed sheets. The fault table is from the makers' own manuals in *09_manuals/*; the symptom guide below pairs each problem with a product you actually have, worked out for this pool. Full detail in *10_products/CATALOGUE.txt*.

WHAT THE TRI PRO SHOWS	WHAT IT MEANS	WHAT TO DO
`No Flow`	Too little or no water through the cell.	Production stops on its own. Check the pump is running and the cell is not scaled. The Flow LED lit with the pump off is normal, not a fault.
`Check Salt`	Salt between 3,000 and 4,000 ppm, depending on water temperature.	Bring it above 4 g/L. One 25 kg sack raises this pool by 0.345 g/L.
`Output Fault`	Possible power supply problem.	Kill the power at the outlet and call someone.
`Cleaning`	The cell is reversing polarity. Not a fault.	Wait about five minutes.
`pH Low` `pH High`	Empty drum, or a probe needing cleaning or calibration. High: a circulation or probe fault.	Check the acid drum, the peristaltic tube, and when the probe was last calibrated.
`pH ERROR`	Dosed a full set of cycles without reaching set point, **or has not run for over 72 hours**.	Note the second half. This one fires from inactivity too, not only from failure.
`ACL Low` `ACL High`	Under- or over-production, or a circulation problem.	Check salt, cell condition and flow before anything else.
`ACL ERROR`	Thirty cumulative hours producing without reaching set point, or thirty hours idle.	Also fires from inactivity.
`pH ---`	pH regulation is not enabled yet. Normal.	It enables itself about eight hours after power-on.

SYMPTOM	CAUSE	WHAT TO REACH FOR, FOR THIS POOL
`Clear but faintly cloudy`	Fine particles passing straight through the sand. This pool's weak point, because the bed runs at a high velocity. **Not tired sand:** it was renewed about September 2025, so age is not the cause and a flocculant will hold.	**Check the gauge first.** If it is near 1.0 the filter is the cause and no chemical will fix it. Otherwise flocculant: 1,088 g pre-diluted into the balance tank, backwash every 4 to 5 hours until clear, then 72.5 g each evening after every backwash. **Also run the robot with its ultra-fine cartridges.** It filters independently of the sand and catches the size of particle the bed is passing, which makes it a standing remedy rather than a dose. Rinse its cartridges after every cycle or it stops picking anything up.
`Cloudy, hard water, visible scale`	Calcium coming out of solution.	Svernante, 725 to 1,450 g. Only once the water is already clear, or it is spent flocculating dirt instead.
`Green, or slippery walls`	Algae.	Algaecide, 1.45 L initial then 363 mL weekly. Expect foam, and expect the cell to work harder while a quat is in the water.

SYMPTOM	CAUSE	WHAT TO REACH FOR, FOR THIS POOL
`Chlorine low though the cell is running`	Either the cell is not producing, or the chlorine is locked up by cyanuric acid.	Work the fault table above first. Then measure cyanuric acid: above 50 mg/L is lock-up, and the target outdoors is 25 to 30. **The cure is dilution, not another chemical.**
`pH keeps climbing`	Normal on a salt pool: electrolysis drives pH up. The acid drum should be handling it.	Check the drum, the peristaltic tube and the probe calibration before dosing by hand. Then pH minus, 290 to 435 g per 0.1 unit.
`Pump will not start`	On an overflow pool the pump draws from the tank, so a **low-level interlock** holds it off when the tank is too low. On `M` you bypass the time clock but not the interlock.	Check the tank level on the clear column first, then the motor thermal reset, the C16 MCB and the RCD. The outlet is at 0.35 m, so there is nothing to draw below that; start at about 0.55 m. **Do not defeat it:** if the column's isolating valve is shut the floats read satisfied and the interlock will permit a dry run.

SYMPTOM	CAUSE	WHAT TO REACH FOR, FOR THIS POOL
`pH too low`	Usually over-correction.	pH plus. **The two tubs are not interchangeable by volume:** 290 to 435 g of the Lapi against 1,450 g of the NortemBio, for the same 0.1 unit.

THE TWO TRAPS IN THE RECUPERO BOTTLE

- It is sodium bromide, and it exists to recover oxidising power once cyanuric acid passes 50 mg/L. A part-used bottle is evidence this pool has had chlorine lock at least once.
- Its own shock step calls for an unstabilised chlorine, calcium hypochlorite. **You do not have any.** Using the granular chlorine you do have would add more cyanuric acid and deepen the very problem you were treating.
- Its target pH is 7.5 to 7.7, above the 7.2 the TRi PRO holds, because bromine chemistry wants the higher figure.
- Sodium bromide in a salt pool turns the cell into a bromine generator, and the bromide stays until it is diluted out. Worth one technician's opinion before reaching for it again.

MAINTENANCE THE SHEETS NEVER MENTIONED

1. **Calibrate the pH and ACL probes at least every two months during the season**, in pH 7.5 buffer. Probes drift, so the pH the unit believes it is holding may not be the pH in the pool.
2. **Press the test button on the 30 mA RCD every six months.** It is the life-safety device in a wet underground room and nothing in the records says it has ever been tested.
3. **Write down the gauge reading straight after every backwash.** That is this filter's clean baseline, and the climb away from it is the real signal that the sand is loading.
4. Move the valve as Kripsol says: motor stopped, press the handle down firmly first because it acts as a clutch, then move it.

Note: the clear level column has its own isolating ball valve at its foot. Closed, the column freezes at whatever level it was showing and the automatic top-up quietly stops working, with nothing else in the room looking wrong.

§6 THE CHEMICAL STOCK

Nine products on the shelf, plus the acid drum in this room. Every one is photographed with its label in *10_products/*, and doses below are the label's own rate applied to 72.5 m^3. *CATALOGUE.txt* in that folder carries the lot numbers, the hazard statements and the working.

PRODUCT	WHAT IT ACTUALLY IS	SIZE	FOR THIS POOL
`pH Più granulare Lapi`	**Sodium carbonate**, CAS 497-19-8. Soda ash.	`25 kg`	290 to 435 g per 0.1 pH up. **Also the correct neutraliser for an acid spill**, applied gradually. Never caustic soda.
`pH+ Increaser NortemBio`	A second, different pH raiser.	`6.5 kg`	1,450 g per 0.1 pH up. Roughly four times the Lapi rate for the same job.
`pH Meno granulare Lapi`	**Sodium bisulfate**, CAS 7681-38-1.	`8 kg`	290 to 435 g per 0.1 pH down. Not the gentle alternative to the acid drum: for the same movement it leaves about 2.15 times the sulphate.
`Cloro granulare 56% Acquaverde`	**Sodium dichloroisocyanurate**, CAS 51580-86-0. Stabilised chlorine.	`5 kg`	Shock 725 to 1,088 g, pre-dissolved into the balance tank. Not for routine use: the cell makes the chlorine. Every dose also adds cyanuric acid.
`Flocculante Klep`	**Polyaluminium chloride**, under 20%.	`5 kg`	Curative 1,088 g, preventive 72.5 g. The right tool for this pool's particular weakness.
`Antialghe Axton`	**Benzalkonium chloride** at 4%, CAS 68424-85-1.	`5 L`	Initial 1.45 L, then 363 mL weekly. Foams, and consumes chlorine.
`Recupero Controlchemi`	**Sodium bromide.**	`5 L, a third full`	Shock 725 g. Read the warning in the

PRODUCT	WHAT IT ACTUALLY IS	SIZE	FOR THIS POOL
			previous section before using it at all.
`Svernante Controlchemi`	**Aminotrimethylene phosphonic acid**, CAS 6419-19-8. A phosphonate scale inhibitor.	`5 kg`	Winter 3.6 L, so about one container a season. Water must be clear first, pumps running 4 to 8 hours before you stop the plant.
`Sale Poseidon Atisale`	Sea salt, no additives, **EN 16401 Quality A**, sold for electrolysis pools. The right salt, and those two marks are why.	`25 kg sacks`	About 12 sacks for a full charge to 4 g/L, which lands at 4.1. One sack raises this pool 0.345 g/L. **Do not go above 5.0 g/L**: that is the robot cleaner's hard limit, and it caps the range well below the 9 g/L the sheets allow.
`Sulphuric acid 36%` *in this room*	The pH minus the TRi PRO doses automatically. Battery electrolyte grade, not a pool product.	`25 kg drum`	About 1 kg per 0.2 of pH reduction. See the safety section.

FOUR THINGS THE SHELF IS MISSING

`CYA test` A cyanuric acid test. Three products on that shelf depend on knowing this number and none of them can tell you it. Without it the Recupero is guesswork.

`shock` An unstabilised chlorine, calcium hypochlorite or similar. Both the Recupero and the Svernante labels call for one and there is none here.

`sulphate` A sulphate test, given years of sulphuric acid on a resin-finished pool with no exposed grout, so this is a low-priority reading rather than the risk it first looked like: it is protecting the balance tank's coating and any place the pool's resin has failed.

`soda ash` Some of the 25 kg of soda ash you already own, kept **in this room** for acid spills. It is simply on the wrong side of the building.

CORRECT THIS ON THE PRINTED SHEET

- The fifth note on the operating sheet says "*Il bidone del cloro*", the chlorine drum. It is not chlorine. The container is **sulfuric acid, ACIDO SOLFORICO 36%**, confirmed from the label. On a salt pool the chlorine is made by the cell; the jug is the pH minus that the TRi PRO doses.
- **Never let anyone add a chlorine product to that jug, or store one beside it.** This is not general caution. Your granular chlorine carries **EUH031 on its own label: contact with acids liberates a toxic gas.** The gas is chlorine. The room is 4.52 m² holding roughly 10 to 11 m³ of air, below ground, with one door. The wording on the sheet is exactly the kind of thing that could cause that mistake, so it is worth correcting the print-out by hand.
- **The white bucket in photograph 34 of *04_plant-room/* is marked UN 3077**, which is the granular chlorine's transport code. Whatever brought it into this room, it should not stand here. Keep the chlorine and the acid in different rooms.
- Handling: H314, causes severe skin burns and eye damage. Gloves and eye protection when changing the drum, and never add water to acid. Keep **sodium bicarbonate or soda ash** down here for spills, applied gradually, and **never caustic soda**, which reacts violently with sulfuric acid. You already own 25 kg of soda ash: it is the pH plus on the shelf.
- **36% is stronger than pool practice recommends.** PWTAG guidance is that sulphuric acid should never be supplied to pools above 25%, and that acids should generally be bought, stored and used below 15%. Separately, sulphuric acid above **15% w/w** is a restricted explosives precursor under EU Regulation 2019/1148, Annex I. That is a reason to ask your supplier, not a verdict from me. The practical answer is to move to a pool-grade pH minus at 15% or below when this drum runs low, and have the drum taken away properly rather than diluted down a drain.
- **Sulphate accumulates, but this pool is less exposed than it first appeared.** Sulphuric acid leaves sulphate behind, and the recognised limit of around **360 mg/L** exists to stop it attacking cementitious grout. **This pool has no grout:** it is finished in smooth black resin, so the mechanism that lifts tiles does not apply. What remains is the balance tank, whose coating is the only thing between the water and its concrete, and anywhere the pool's resin has failed and exposed the shell. Still worth one reading because nobody has ever taken one, but it drops a long way down the list. An earlier version of this document treated this as a tiled pool and rated it higher; that was my inference and it was wrong. The granular pH minus is still worse on this count than the acid, for the same pH movement.

BECAUSE THE ROOM IS BELOW GROUND

- **The sump pump is the only drain.** Nothing leaves that floor by gravity. If it has failed, the first you will know is standing water over the pump motor and the panel. Worth running it deliberately at the start of each season rather than finding out.
- **Ventilation matters more here.** A 25 kg drum of sulfuric acid in a confined space below ground is a different proposition to the same drum in an airy outbuilding. Check the room has a working vent, and open the door for a while before working in there.
- **Salt electrolysis makes hydrogen** as a by-product. Normally it is carried away dissolved in the water, which is one more reason never to run the cell without flow. Isolate it on B and C whenever the pump is off for any length of time.

IF WATER IS COMING IN: WHAT TO SHUT, IN ORDER

- The pool sits at ground level and this room is underneath it. That means the water upstairs is permanently trying to get down here, and a burst or an open joint will keep flowing long after the pump stops. **Switching the pump off does not stop it.**
- Shut, in this order: 7 (balance tank outlet), then L (pool floor discharge), then all four D returns. That cuts every path between the pool and this room. G as well if you can reach it.
- Then the sump pump N has a chance of keeping up. It is the only drain the room has.
- For a sense of scale: **the balance tank holds 3.69 m³ in normal operation, which over this floor stands 0.82 m deep**, and 1.21 m if its safety overflow were blocked and it had filled to the coating line. The pool overhead holds roughly six times this room's entire air volume.
- This sequence is mine and is not on the sheets: it follows from the levels rather than from anything written down. It is worth checking with a technician, and worth knowing before you need it.

ITEM	DETAIL	SOURCE
Pump	DAB EuroPro 200 M, no. 2.1213, TF 60 S1. Q 31-29-26 m^3/h at H 3-10-12 m, Hmin 5 m, max 2.5 bar. P2 2 HP, P1 1.9 kW. 220-240 V, 8.6 A, 2800 rpm, 50 Hz, class F, IP 55, 40 µF 450 V. App. no. NSW25167.	Nameplate
Filter valve	Kripsol side-mount 6-way, 7 handle positions (0 and 1 to 6), listed in the procedures section.	Valve body and sheets
Sand filter	**Sand renewed about September 2025**, so roughly one year into a five to seven year life. Recorded as new sand rather than washed sand; those are different clocks. Kripsol Ø760, Serie 12. Filtration area 0.45 m^2, rated flow **22 m^3/h**, total sand charge **225 kg**, sand grading **0.4 to 0.8 mm**. Working pressure 0.5 to 1.5 kg/cm^2, test pressure 3.5 kg/cm^2, design filtration rate 50 m^3/h per m^2. Kripsol, Ugena (Toledo), Spain.	Vessel plate
Flow, worth a look	The filter is rated 22 m^3/h at 50 m^3/h per m^2 of sand. The pump is rated 31, 29 and 26 m^3/h at 3, 10 and 12 m of head. Read on their own those figures say the pump over-flows the filter, but the pump nameplate is the wrong number to use: what actually flows is set by the narrowest part of the circuit, and the tank draws through a single modest-bore line. At a sane suction velocity of about 2 m/s a 63 mm line carries roughly 18 m^3/h and a 75 mm line roughly 26. **So on 63 mm you are under the filter's rating and the pump is simply oversized, which costs electricity and bearing life rather than damaging the filter. On 75 mm you are at the plate limit.** Measuring the suction pipe's outside diameter settles which. For reference, the DIN 19643 figure for single-layer sand is 30 m^3/h per m^2, well below what this filter's own plate allows.	Plates, plus manufacturer data
Chlorinator	Zodiac TRi PRO, salt electrolysis with pH and ORP regulation and an integral peristaltic dosing pump. Cell in a clear housing on a three-valve arrangement (B, C, A).	Front panel
Panel	Tele-Matic enclosure. Hager CDC 724H 25 A 30 mA RCD, MJN516A C16 MCB, LS501 10.3×38 fuse holders, ST314 40 VA 230 to 12/24 V auxiliary transformer, 24 h time clock. Two Lovato LVM 20 and LVM 25 level relays for conductive liquids.	Readable in photos

ITEM	DETAIL	SOURCE
`Top–up`	Mains through tap E, a strainer, and solenoid F driven by the level relays. A manual bypass loop with a third blue valve lets you fill by hand if the solenoid fails. Discharges into the **return manifold**, not the tank. Whether the line carries a non-return valve is unverified, and it should: the pool sits above this room, so pool water could siphon toward the supply if mains pressure dropped.	Observed, plus the discharge point reported on site
`pH correction`	ACIDO SOLFORICO 36% (sulfuric acid), 25 kg drum, Siac srl, Torino. UN 2796, ADR class 8 packing group II, index 016-020-00-8, H314. Composition line reads "elettrolito per batterie ... acido solforico 30/31 Bé", so it is battery-electrolyte grade rather than a pool-branded product. Dosed automatically by the TRi PRO peristaltic pump. Sheet figure: about 1 kg per 0.2 of pH reduction.	Drum label
`Robot cleaner`	**Maytronics Dolphin Maximus X70**, fitted with **soft Wonder foam brushes**, changed from the PVC ones in September 2026. That is the correct pairing: Maytronics specify foam for smooth surfaces and PVC for rough pebble or quartz render, and this pool is smooth resin. Cleans floor, walls and waterline on a 2.5 hour cycle. 17 m cable, rated for in-ground pools up to 15 m, so this 13 m pool is inside its range but not by much. 11.17 kg, supplied with a caddy, an ultra-fine filter kit and Wi-Fi scheduling through the MyDolphin Plus app. Four year warranty, so keep the proof of purchase. Water limits: **chlorine max 4 mg/L, pH 7.0 to 7.8, 6 to 35 °C, and salt max 5.0 g/L**. Power supply is switch-mode IP54, 100-250 V, 180 W, and must sit **at least 3.5 m from the pool edge and 11 cm off the ground**, roughly mid-way along the long side, with no extension cord. Its manual is badged *CLASSIC 8+ / TOP 8* and never says Maximus X70; it is nonetheless the right document. **It is completely independent of the plant room:** its own pump, its own filter cartridges, no connection to the circuit, which means it does not use the manual vacuum H or the vacuuming procedure in the procedures section at all.	Maytronics manual 8180035, in *09_manuals/*

ITEM	DETAIL	SOURCE
Sump	N, a floor gully with a submersible pump. The room is below ground, so the floor cannot drain by gravity and this pump is the only way water gets out. That makes it safety equipment rather than a convenience.	Sheets, gully visible
Turnover	At the filter's rated 22 m^3/h, a 70 to 75 m^3 pool turns over in roughly 3.2 to 3.4 hours. Overflow pools are normally run faster than skimmer pools, so that is a sensible figure. Actual flow depends on the real head and the state of the sand.	Arithmetic

§9 WHAT IS STILL OPEN

Nothing in this list is known and simply unrecorded. It is the honest inventory of what this document does not know, kept here so it can be closed off later rather than quietly forgotten. The third column of every row says what closing it would change, so you can see at a glance which are worth a trip to the plant room and which are only tidiness.

MEASURE THIS	WHERE	WHAT IT CHANGES
`Suction pipe outside diameter`	The low pipe running from the compensation valve to the pump. 50, 63, 75 or 90 mm.	Whether the filter is being over-flowed at all. On 63 mm it is not, and the pump is merely oversized, which costs electricity rather than sand. On 75 mm it sits at the plate limit. The equipment section currently states both cases because this is unmeasured.
`Is there a second valve?`	Follow the pipe from the compensation valve behind the D drops toward the pump. Look for a second red handle.	Whether G and 7 are one valve or two. The sheets give both the same job and only one appears in any photograph. **This is the instruction you would be following with water coming in**, so it matters more than its size suggests.
`Where the acid injects`	Confirmed on site 2026-09-08: a small plastic tube runs from the acid drum to the TRi PRO, so the drum feeds that module's peristaltic pump. The tube still unseen is the other one, the one leaving the TRi PRO. Follow it and note which pipe it enters, and whether that point is before or after the cell.	Whether acid ever meets the cell undiluted. Downstream of the cell is the normal arrangement and is what this document assumes; upstream would put concentrated acid across the plates. It also tells you where to look first for a leak, because that tube carries 36% sulfuric acid.
`Ceiling height`	This room, floor to ceiling. The 3D scan puts a floor under it: its vertex density stops at 2.33 m, so the ceiling is **at least** that. It found no ceiling plane, because the phone was never pointed up.	The room's air volume, and with it whether the acid-storage warning in the safety notes is a measured fact or my judgement. Now known to be **10.5 m³ or more.** A tape would close it.
`Column mark spacing`	The clear level column, distance between the marks 1 to 4.	Precision only. Multiply by 27.3 and the column reads in litres instead of positions.

MEASURE THIS	WHERE	WHAT IT CHANGES
`Wall thickness`	Between this room and the tank.	Precision only, and probably nothing: the 7 cm on the scaled plan is almost certainly a drawn separation. Worth confirming, because 5.46 m^3 of water sits on the other side.
`What the resin actually is`	The product name or datasheet for the black resin coating, from whoever applied it.	Its chlorine, pH and temperature tolerances. It matters before the commissioning shock dose, which will put roughly 8 mg/L of free chlorine into the water: most coatings take that comfortably, but you would want to know rather than assume. It is also what you would need if a patch ever has to be repaired.

TEST THIS	WHY	THRESHOLD
`Cyanuric acid`	The most consequential unknown in the whole chemistry. Three products on the shelf depend on this number and none of them can measure it. A part-used bottle of sodium bromide is evidence this pool has hit chlorine lock at least once, and above the threshold the cell's output is simply being wasted.	`above 50 mg/L is lock-up` `target 25 to 30 outdoors`
`Sulphate`	Years of sulphuric acid dosing, never measured. Lower priority than it looked: the pool is resin-finished with no grout, so the limit is only protecting the balance tank's coating and any place the pool's resin has failed. Cheap enough to do once.	`360 mg/L`
`A pH and chlorine test kit`	Do you own one? Never established.	Every target in the water section is unusable without one: pH 7.2 to 7.4, free chlorine 0.6 to 1.5, salt 4.0 to 5.0. Right now the holding doses during the refill are going in blind. **Cheaper and more useful than either test above.**
`Non-return valve on the top-up`	Whether the mains feed into the return manifold has one.	A drinking-water question rather than a pool one. The pool sits above this room, so if mains pressure ever dropped, pool water could siphon

TEST THIS	WHY	THRESHOLD
		back toward the supply. I cannot tell from the photographs whether there is one.

FIVE SMALL THINGS, NONE OF THEM EXPENSIVE

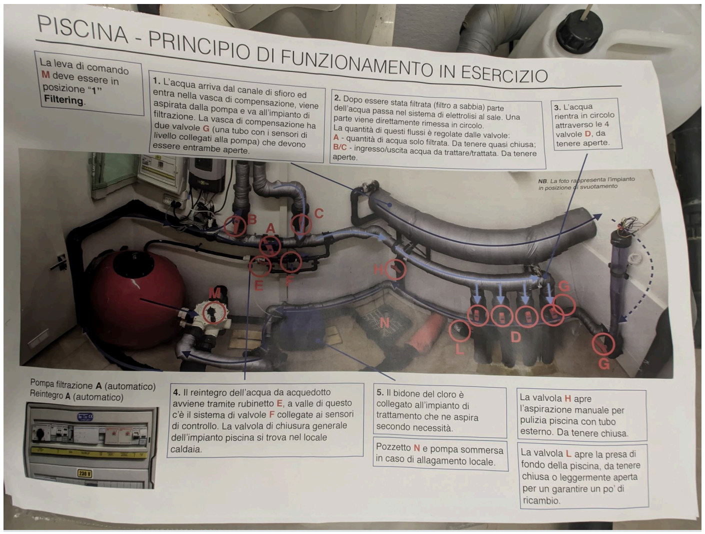
1. **Move some soda ash into this room.** You own 25 kg of it as the pH plus on the shelf. It is the correct neutraliser for an acid spill and it is currently on the wrong side of the building.
2. **Keep the chlorine out of this room, permanently.** Its own label carries EUH031: contact with acids liberates a toxic gas.
3. **Clip the overflow pipe's open end** in the balance tank. It lies loose on the floor and will move when the pool spills hard.
4. **Look at the rust staining and blown plaster** below the hatch in the service corridor. Water has been coming through or condensing there.
5. **Buy an unstabilised chlorine** for shock dosing, calcium hypochlorite or similar. Both the Recupero and the Svernante labels call for one and there is none on the shelf.

Note: everything above that needs the balance tank wants doing while it is empty and open. That will not stay true.

ONE THING NOBODY CAN MEASURE, ONLY FIND OUT

`dilution` **Has the pool ever been partly drained and refilled?** Backwash losses aside, nothing in the record says so. It bears directly on both figures in the table above: cyanuric acid and sulphate both accumulate and both are only removed by dilution.

The four printed pages that live in the room. Everything in the procedures, the water figures and the safety notes above is transcribed and translated from these. Where a maker's manual contradicts them, that is said at the point it matters: the filter gauge is the one case where this document now follows the manual instead of the sheet. Kept here at full readability so the original is never more than one file away.



PISCINA - PRINCIPIO DI FUNZIONAMENTO IN ESERCIZIO

La leva di comando M deve essere in posizione "1" **Filtering**.

1. L'acqua arriva dal canale di sfioro ed entra nella vasca di compensazione, viene aspirata dalla pompa e va all'impianto di filtrazione. La vasca di compensazione ha due valvole G (una tubo con i sensori di livello collegati alla pompa) che devono essere entrambe aperte.

2. Dopo essere stata filtrata (filtro a sabbia) parte dell'acqua passa nel sistema di elettrolisi al sale. Una parte viene direttamente rimessa in circolo. La quantità di questi flussi è regolate dalle valvole:
A - quantità di acqua solo filtrata. Da tenere quasi chiusa;
B/C - ingresso/uscita acqua da trattare/trattata. Da tenere aperte.

3. L'acqua rientra in circolo attraverso le 4 valvole D, da tenere aperte.

NB. La foto rappresenta l'impianto in posizione di svuotamento

Pompa filtrazione **A** (automatico)
Reintegro **A** (automatico)

4. Il reintegro dell'acqua da acquedotto avviene tramite rubinetto E, a valle di questo c'è il sistema di valvole F collegate ai sensori di controllo. La valvola di chiusura generale dell'impianto piscina si trova nel locale caldaia.

5. Il bidone del cloro è collegato all'impianto di trattamento che ne aspira secondo necessità.

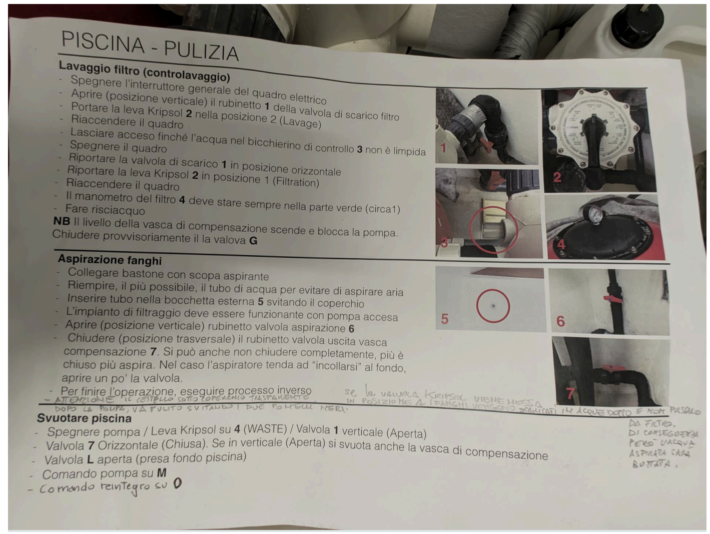
Pozzetto N e pompa sommersa in caso di allagamento locale.

La valvola H apre l'aspirazione manuale per pulizia piscina con tubo esterno. Da tenere chiusa.

La valvola L apre la presa di fondo della piscina, da tenere chiusa o leggermente aperta per un garantire un po' di ricambio.

PRINCIPIO DI FUNZIONAMENTO

How the system runs. The source of the letters A, B, C, D, E, F, G, H, L, M and the sump N.



PISCINA - PULIZIA

Lavaggio filtro (controlavaggio)
- Spegnere l'interruttore generale del quadro elettrico
- Aprire (posizione verticale) il rubinetto **1** della valvola di scarico filtro
- Portare la leva Kripsol **2** nella posizione 2 (Lavage)
- Riaccendere il quadro
- Lasciare acceso finché l'acqua nel bicchierino di controllo **3** non è limpida
- Spegnere il quadro
- Riportare la valvola di scarico **1** in posizione orizzontale
- Riportare la leva Kripsol **2** in posizione 1 (Filtration)
- Riaccendere il quadro
- Il manometro del filtro **4** deve stare sempre nella parte verde (circa1)
- Fare risciacquo

NB Il livello della vasca di compensazione scende e blocca la pompa. Chiudere provvisoriamente il la valova **G**

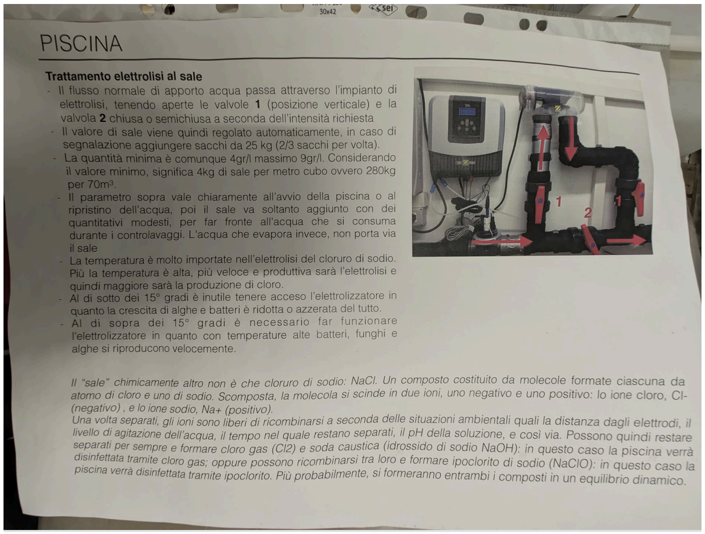
Aspirazione fanghi
- Collegare bastone con scopa aspirante
- Riempire, il più possibile, il tubo di acqua per evitare di aspirare aria
- Inserire tubo nella bocchetta esterna **5** svitando il coperchio
- L'impianto di filtraggio deve essere funzionante con pompa accesa
- Aprire (posizione verticale) rubinetto valvola aspirazione **6**
- Chiudere (posizione trasversale) il rubinetto valvola uscita vasca compensazione **7**. Si può anche non chiudere completamente, più è chiuso più aspira. Nel caso l'aspiratore tenda ad "incollarsi" al fondo, aprire un po' la valvola.
- Per finire l'operazione, eseguire processo inverso
- ATTENZIONE IL CESTELLO SOTTO COPERCHIO TRASPARENTE DOPO LA POMPA, VA PULITO SVITANDO I DUE POMELLI NERI.

SE LA VALVOLA KRIPSOL VIENE MESSA IN POSIZIONE 4 I FANGHI VENGONO SCARICATI IN ACQUEDOTTO E NON PASSANO DA FILTRO. DI CONSEGUENZA PERÒ L'ACQUA ASPIRATA SARÀ BUTTATA.

Svuotare piscina
- *Spegnere pompa / Leva Kripsol su* **4** (WASTE) / *Valvola* **1** verticale (Aperta)
- *Valvola* **7** *Orizzontale (Chiusa). Se in verticale (Aperta) si svuota anche la vasca di compensazione*
- *Valvola* **L** *aperta (presa fondo piscina)*
- *Comando pompa su* **M**
- Comando reintegro su 0

PULIZIA

Backwash, vacuuming and emptying, with handwritten notes in the margins.



PISCINA

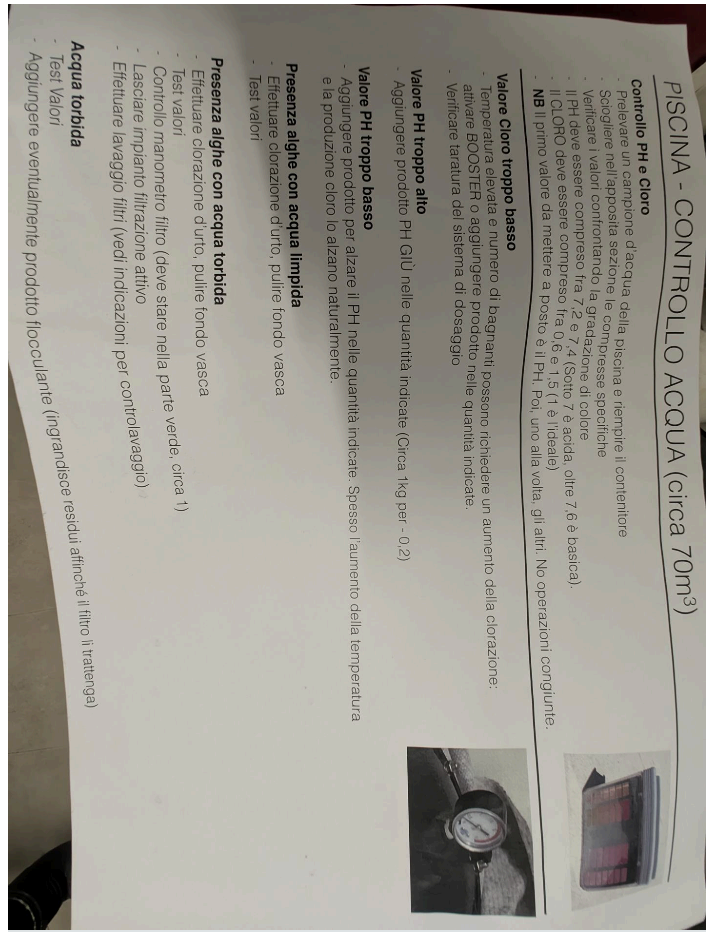
Trattamento elettrolisi al sale

- Il flusso normale di apporto acqua passa attraverso l'impianto di elettrolisi, tenendo aperte le valvole **1** (posizione verticale) e la valvola **2** chiusa o semichiusa a seconda dell'intensità richiesta
- Il valore di sale viene quindi regolato automaticamente, in caso di segnalazione aggiungere sacchi da 25 kg (2/3 sacchi per volta).
- La quantità minima è comunque 4gr/l massimo 9gr/l. Considerando il valore minimo, significa 4kg di sale per metro cubo ovvero 280kg per $70m^3$.
- Il parametro sopra vale chiaramente all'avvio della piscina o al ripristino dell'acqua, poi il sale va soltanto aggiunto con dei quantitativi modesti, per far fronte all'acqua che si consuma durante i controlavaggi. L'acqua che evapora invece, non porta via il sale
- La temperatura è molto importate nell'elettrolisi del cloruro di sodio. Più la temperatura è alta, più veloce e produttiva sarà l'elettrolisi e quindi maggiore sarà la produzione di cloro.
- Al di sotto dei 15° gradi è inutile tenere acceso l'elettrolizzatore in quanto la crescita di alghe e batteri è ridotta o azzerata del tutto.
- Al di sopra dei 15° gradi è necessario far funzionare l'elettrolizzatore in quanto con temperature alte batteri, funghi e alghe si riproducono velocemente.

Il "sale" chimicamente altro non è che cloruro di sodio: NaCl. Un composto costituito da molecole formate ciascuna da atomo di cloro e uno di sodio. Scomposta, la molecola si scinde in due ioni, uno negativo e uno positivo: lo ione cloro, Cl- (negativo) , e lo ione sodio, Na+ (positivo).
Una volta separati, gli ioni sono liberi di ricombinarsi a seconda delle situazioni ambientali quali la distanza dagli elettrodi, il livello di agitazione dell'acqua, il tempo nel quale restano separati, il pH della soluzione, e così via. Possono quindi restare separati per sempre e formare cloro gas (Cl2) e soda caustica (idrossido di sodio NaOH): in questo caso la piscina verrà disinfettata tramite cloro gas; oppure possono ricombinarsi tra loro e formare ipoclorito di sodio (NaClO): in questo caso la piscina verrà disinfettata tramite ipoclorito. Più probabilmente, si formeranno entrambi i composti in un equilibrio dinamico.

TRATTAMENTO ELETTROLISI AL SALE

Salt levels, temperature thresholds, and how the electrolysis works.



PISCINA - CONTROLLO ACQUA (circa $70m^3$)

Controllo PH e Cloro
- Prelevare un campione d'acqua della piscina e riempire il contenitore
- Sciogliere nell'apposita sezione le compresse specifiche
- Verificare i valori confrontando la gradazione di colore
- Il PH deve essere compreso fra 7,2 e 7,4 (Sotto 7 è acida, oltre 7,6 è basica).
- Il CLORO deve essere compreso fra 0,6 e 1,5 (1 è l'ideale)
- **NB** Il primo valore da mettere a posto è il PH. Poi, uno alla volta, gli altri. No operazioni congiunte.

Valore Cloro troppo basso
- Temperatura elevata e numero di bagnanti possono richiedere un aumento della clorazione: attivare BOOSTER o aggiungere prodotto nelle quantità indicate.
- Verificare taratura del sistema di dosaggio

Valore PH troppo alto
- Aggiungere prodotto PH GIÙ nelle quantità indicate (Circa 1kg per - 0,2)

Valore PH troppo basso
- Aggiungere prodotto per alzare il PH nelle quantità indicate. Spesso l'aumento della temperatura e la produzione cloro lo alzano naturalmente.

Presenza alghe con acqua limpida
- Effettuare clorazione d'urto, pulire fondo vasca
- Test valori

Presenza alghe con acqua torbida
- Effettuare clorazione d'urto, pulire fondo vasca
- Test valori
- Controllo manometro filtro (deve stare nella parte verde, circa 1)
- Lasciare impianto filtrazione attivo
- Effettuare lavaggio filtri (vedi indicazioni per controlavaggio)

Acqua torbida
- Test Valori
- Aggiungere eventualmente prodotto flocculante (ingrandisce residui affinché il filtro li trattenga)

CONTROLLO ACQUA

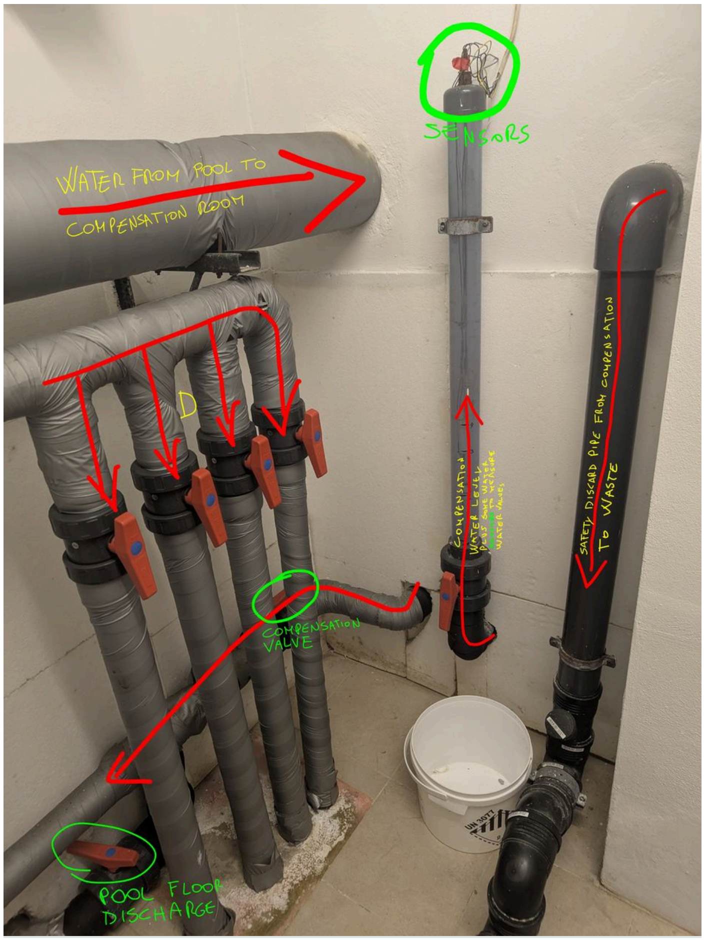
pH and chlorine targets and what to do when they drift.

§11 REFERENCE PHOTOGRAPHS

The images every factual claim in this document rests on. If anything here ever looks wrong, these are what to check it against.

THE ROOM, 216 DEGREES

Phone photosphere, 39 megapixels. The best single record of the room as it stands.

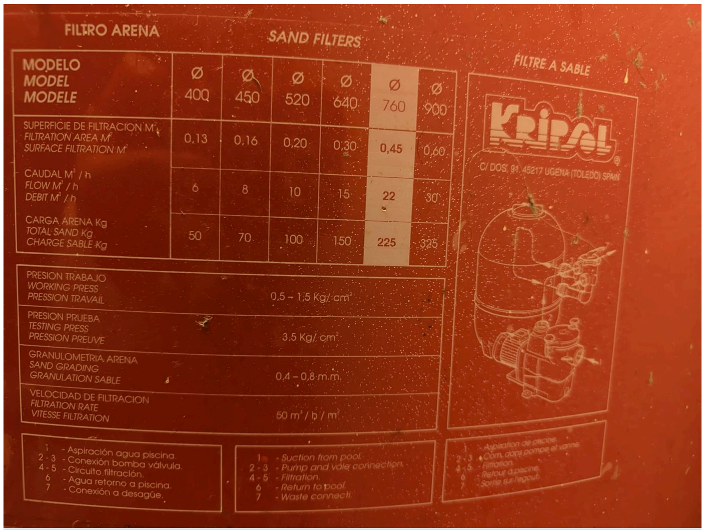


THE MANIFOLD WALL, YOUR NOTES

Your own annotations: the pool overflow heading to the compensation room, the sensors, the compensation valve, the safety discharge, and the pool floor discharge.

THE ELECTRICAL PANEL

Every label readable. Hager gear, two Lovato level relays, the 24 hour time clock.

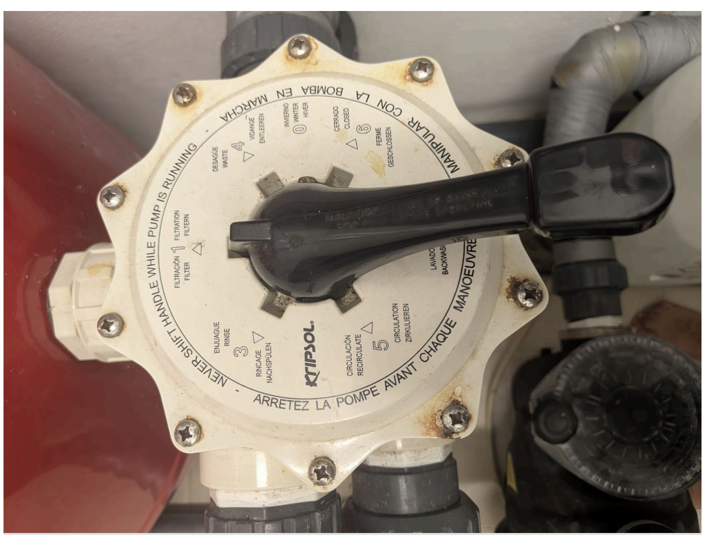


FILTER DATA PLATE

Kripsol. The 760 column is the highlighted one: 0.45 square metres, 22 cubic metres an hour, 225 kg of sand, 0.4 to 0.8 mm.

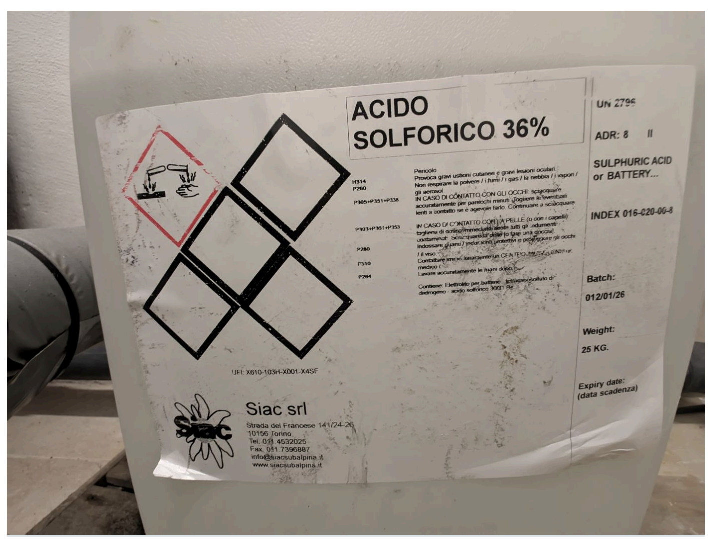
PUMP NAMEPLATE

DAB EuroPro 200 M. 2 HP, 1.9 kW, 31 to 26 cubic metres an hour, 2800 rpm, IP55.



KRIPSOL VALVE DIAL

All seven positions, and the warning cast into the rim in four languages.



THE DRUM LABEL

ACIDO SOLFORICO 36 percent. UN 2796, class 8, H314. Sold as battery electrolyte.

FOLDER	WHAT IT HOLDS
`00_inbox/`	Staging. New photographs and documents go in here named however they arrived, and are filed and renamed from there.
`01_survey/`	This document as HTML and PDF, plus the diagram as a PNG. The HTML is entirely self-contained: no internet, no linked files, every photograph embedded in it.
`02_printed-sheets/`	The four printed pages kept in the room, photographed. Everything in the legend, the procedures and the water figures is transcribed and translated from these.
`03_nameplates/`	The five plates and labels every specification rests on: the Kripsol filter plate, the DAB pump nameplate, the valve dial, and the acid drum.
`04_plant-room/`	Thirty-four photographs of this room, numbered in the order they were taken. Number 34 is the manifold wall with your own annotations on it.
`05_balance-room/`	Six photographs of the balance tank next door and the hatch that reaches it, with *ANNOTATION-KEY.txt* recording what the coloured marks on photograph 04 mean.
`06_drawings/`	Your own drawings: the dimensioned sketch of this room, and the scaled plan showing both rooms and how they sit together. The plan is genuinely to scale, checked at 296 pixels per metre against your measurements.
`07_video/`	Three phone pans: two 360 degree sweeps and one floor-level survey.
`08_panoramas/`	Four flat, straightened wall views, each reprojected from the 39-megapixel photosphere and **safe to count from**: photograph 07 has been checked at four drops. The numbering starts at 05 because 01 to 03 were stitched images that drew six drops where there are four, and were deleted. The folder's *READ-ME.txt* keeps that lesson.
`09_manuals/`	The makers' own manuals for the filter, the chlorinator, the TRi PRO module and the pump, with *SOURCES.txt* giving the official link for each and the figures taken from it.
`10_products/`	The nine chemicals on the shelf, each photographed with its label, grouped by what they do. *CATALOGUE.txt* gives the active ingredient, the dose worked out for this pool, the hazards worst first, and a symptom-to-action guide.
`11_scan/`	A 3D scan of this room, 1.29 million triangles. Its geometry independently confirms the tape measure: 2.52 m against 2.55 along the long axis, 2.00 against

FOLDER	WHAT IT HOLDS
	1.96 across. Not usable for counting anything, because thin pipework does not survive a scan at that density.

Measured dimensions for both rooms, and what follows from them, are in *DIMENSIONS.txt*. What every chemical is and what to reach for is in *10_products/CATALOGUE.txt*. Every file is named to one rule, set out in *NAMING.txt* at the top of the folder: an index, then what the file shows, then an optional note on how it differs from its neighbour, then a marked technical token. For proof of what is installed, the 39-megapixel photosphere and the panel photograph are the record. This document is an interpretation of them.

Compiled 2026-09-06 from photographs of the room and from the printed sheets kept in it. The sheets are a household reference rather than a professional specification, and do not necessarily capture everything about this installation. Nothing here has been verified by a physical inspection or against an installation drawing. For evidence of what is installed, the 39-megapixel photosphere and the panel photograph are the record.